

NAME

ptpd2.conf - Precision Time Protocol daemon config file

CONFIGURATION FILE FORMAT

Settings in the PTPd configuration file are grouped into sections and take the form of `section:key="value"` variables. The configuration file can either be formatted that way (preferred) or in .ini file style where a series of `key="value"` variables is grouped into sections using `[section]` headers. Every setting listed here can also be specified as a command line parameter (`--section:key=value`). Quotation marks are optional. **NOTE:** the configuration file must end with a newline.

RELOADING CONFIGURATION

Only a small number of configuration file settings (SNMP, lock file configuration) requires a restart of the PTPd process to take effect. All other settings can be changed while ptpd is running - configuration file is reloaded and checked for changes when PTPd receives the SIGHUP signal. When reloading configuration, PTPd will always attempt to test settings before applying them and once running, will never exit as a result of configuration errors. If it does exit during config refresh, this is most likely a bug.

PRIORITY

Any setting passed as a command line parameter will always take priority over the configuration file, so once ptpd is running, those settings cannot be changed - a warning will be logged on every attempt to change those settings using the configuration file.

CONFIGURATION SECTIONS

ptpengine

PTP protocol specific configuration

clock

Clock related settings

servo

Clock control PI servo configuration

global

Global configuration - logging, etc.

ntpengine

NTP control configuration

CONFIGURATION VARIABLES

`ptpengine:interface` [*STRING*]

usage

Network interface to use - eth0, igb0 etc. (**required**).

default

[none]



ptpengine:backup_interface [*STRING*]

usage

Backup network interface to use - eth0, igb0 etc. When no GM available, slave will keep alternating between primary and secondary until a GM is found.

default

[none]

ptpengine:preset [*SELECT*]

options

none slaveonly masteronly masterslave

usage

PTP engine preset:

none

Defaults, no clock class restrictions

slaveonly

Slave only (clock class 255 only)

masteronly

Master, passive when not best master (clock class 0..127)

masterslave

Full IEEE 1588 implementation: Master, slave when not best master (clock class 128..254)

default

slaveonly

NOTE:

Presets affect the following settings:
ptpengine:slave_only, *clock_no_adjust* and
ptpengine:clock_class (range and default value). To see all preset settings, run `ptpd2 -H (--long-help)`

ptpengine:transport [*SELECT*]

options

ipv4 ethernet

usage

Transport type for PTP packets. **NOTE:** Ethernet transport requires building with *libpcap* and is not supported on Solaris as of 2.3.1, and cannot be enabled on those systems unless `ptpd` is compiled with `--enable-experimental-options`.

default

ipv4

ptpengine:dot2as [*BOOLEAN*]

usage

Enable IEEE 802.1AS / AVB compatibility (transportSpecific field in PTP message headers). Requires Ethernet transport as this is the only mapping used by 802.1AS that PTP supports

default

N

ptpengine:ip_mode [*SELECT*]

options

multicast unicast hybrid

usage

IP transmission mode (requires IP transport):

multicast

uses multicast for all messages

hybrid

uses multicast for sync and announce, and unicast for delay request and response

unicast

uses unicast for all transmission. When unicast mode is selected, destination IP(s) (*ptpengine:unicast_destinations*) must be configured depending on unicast negotiation setting (*ptpengine:unicast_negotiation*) and master or slave role (see: *ptpengine:unicast_destinations*)

default

multicast

ptpengine:unicast_negotiation [BOOLEAN]

usage

Enable unicast negotiation support using signaling messages - as used by the Telecom profile (ITU-T G.8265.1).

default

N

ptpengine:unicast_grant_duration [INT: 30 .. 604800]

usage

Duration (seconds) for which the transmission of unicast messages is granted by a master, or requested by a slave when unicast negotiation is used (*ptpengine:unicast_negotiation*). When using PTPd with other PTP implementations, PTPd will never refuse to grant a message based on the requested duration: it will grant for 30 seconds if requested for any less than 30 seconds, and will grant for 7 days (604800) if requested for any longer.

default

300

ptpengine:disable_bmca [BOOLEAN]

usage

Disable Best Master Clock Algorithm for unicast masters. Only effective for masteronly preset - all Announce messages will be ignored and the clock will transition directly into MASTER state and remain an active master. This behaviour is required for Telecom profile operation.

default

N

ptpengine:use_libpcap [BOOLEAN]

usage

Use libpcap for sending and receiving traffic (automatically enabled in Ethernet mode). Requires building with libpcap - builds made with **--disable-pcap** cannot use this feature, and as of 2.3.1, Solaris systems will not attempt to use libpcap unless compiled with **--enable-experimental-options**

default

N

ptpengine:delay_mechanism [*SELECT*]

options

E2E P2P DELAY_DISABLED

usage

Delay detection mode used - use DELAY_DISABLED for syntonisation only (no synchronisation).

default

E2E

ptpengine:domain [*INT: 0 .. 127*]

usage

PTP domain number.

default

0

ptpengine:any_domain [*BOOLEAN*]

usage

Usability extension: if enabled, a slave-only clock will accept masters from any domain, while preferring the configured domain, and preferring lower domain number. This option should be used for slave-only clocks and should not be used with unicast negotiation. **NOTE:** this behaviour is not part of the standard.

default

N

ptpengine:port_number [*INT: 1 .. 65534*]

usage

PTP port number (part of PTP Port Identity - not UDP port). For ordinary clocks (single port), the default should be used, but when running multiple instances to simulate a boundary clock, The port number can be changed.

default

1

ptpengine:slave_only [*BOOLEAN*]

usage

Slave only mode (sets clock class to 255, overriding value from preset).

default

Y

ptpengine:inbound_latency [*INT*]

usage

Specify latency correction (nanoseconds) for incoming packets.

default

0

ptpengine:outbound_latency [INT]

usage
Specify latency correction (nanoseconds) for outgoing packets.

default
0

ptpengine:offset_shift [INT]

usage
Apply an arbitrary shift (nanoseconds) to offset from master when in slave state. Value can be positive or negative - useful for correcting for antenna latencies, delay asymmetry and IP stack latencies. This will not be visible in the offset from master value - only in the resulting clock correction.

default
0

ptpengine:always_respect_utc_offset [BOOLEAN]

usage
Compatibility option: In slave state, always respect UTC offset announced by best master, even if the the currentUtcOffsetValid flag is announced FALSE. **NOTE:** this behaviour is not part of the standard.

default
N

ptpengine:prefer_utc_offset_valid [BOOLEAN]

usage
Compatibility extension to BMC algorithm: when enabled, BMC for both master and slave clocks will prefer masters announcing currentUtcOffsetValid as TRUE. **NOTE:** this behaviour is not part of the standard.

default
N

ptpengine:require_utc_offset_valid [BOOLEAN]

usage
Compatibility option: when enabled, ptpd2 will ignore Announce messages from masters announcing currentUtcOffsetValid as FALSE. **NOTE:** this behaviour is not part of the standard.

default
N

ptpengine:log_announce_interval [INT: -4 .. 7]

usage
PTP announce message interval in master state. When using unicast negotiation (*ptpengine:unicast_negotiation*), for slaves this is the initial (minimum) interval requested and

for masters this is the minimum interval granted.
(expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.)

default

1

ptpengine:log_announce_interval_max [INT: -1 .. 7]

usage

When using unicast negotiation
(*ptpengine:unicast_negotiation*), this is the maximum
announce interval granted by a master, and the maximum
interval a slave will attempt to request. (expressed as log
2 i.e. -1=0.5s, 0=1s, 1=2s etc.)

default

5

ptpengine:announce_receipt_timeout [INT: 2 .. 255]

usage

PTP announce receipt timeout announced in master state.

default

6

ptpengine:announce_receipt_grace_period [INT: 0 .. 20]

usage

PTP announce receipt timeout grace period in slave state:
when announce receipt timeout occurs, disqualify current
best GM,
then wait n times announce receipt timeout before
resetting. Allows for a seamless GM failover when standby
GMs are slow
to react. When set to 0, this option is not used.

default

0

ptpengine:log_sync_interval [INT: -7 .. 7]

usage

PTP sync message interval in master state. When using
unicast negotiation (*ptpengine:unicast_negotiation*), for
slaves this is the initial (minimum) interval requested and
for masters this is the minimum interval granted.
(expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.)

default

0

ptpengine:log_sync_interval_max [INT: -1 .. 7]

usage

When using unicast negotiation
(*ptpengine:unicast_negotiation*), this is the maximum sync
interval granted by a master, and the maximum interval a
slave will attempt to request. (expressed as log 2 i.e.
-1=0.5s, 0=1s, 1=2s etc.)

default

5



ptpengine:log_delayreq_override [BOOLEAN]

usage

Override the Delay Request interval provided by best master.

default

N

ptpengine:log_delayreq_auto [BOOLEAN]

usage

Automatically override the Delay Request interval (with *ptpengine:log_delayreq_interval*) if the received value is 127 (0X7F), such as in unicast messages, unless using unicast negotiation (*ptpengine:unicast_negotiation*)

default

Y

ptpengine:log_delayreq_interval_initial [INT: -7 .. 7]

usage

Delay request interval used before receiving first delay response (expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.)

default

0

ptpengine:log_delayreq_interval [INT: -7 .. 7]

usage

Minimum delay request interval announced when in master state, in slave state overrides the master interval. (expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.). When using unicast negotiation (*ptpengine:unicast_negotiation*), for slaves this is the initial (minimum) interval requested and for masters this is the minimum interval granted.

default

0

ptpengine:log_delayreq_interval_max [INT: -1 .. 7]

usage

When using unicast negotiation (*ptpengine:unicast_negotiation*), this is the maximum delay request interval granted by a master, and the maximum interval a slave will attempt to request. (expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.).

default

5

ptpengine:log_peer_delayreq_interval [INT: -7 .. 7]

usage

Minimum peer delay request message interval in peer to peer delay mode (expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.). When using unicast negotiation (*ptpengine:unicast_negotiation*), this is the initial (minimum) interval requested by a node from its peer and this is the minimum interval granted for a peer.

default

ptpengine:log_peer_delayreq_interval_max [*INT*: -1 .. 7]

usage

When using unicast negotiation (*ptpengine:unicast_negotiation*), this is the maximum peer delay request interval granted by a node, and the maximum interval a node will attempt to request from its peer. (expressed as log 2 i.e. -1=0.5s, 0=1s, 1=2s etc.).

default

5

ptpengine:foreignrecord_capacity [*INT*: 5 .. 10]

usage

Foreign master record size (Maximum number of foreign masters).

default

5

ptpengine:ptp_allan_variance [*INT*: 0 .. 65535]

usage

Specify Allan variance announced in master state.

default

28768

ptpengine:ptp_clock_accuracy [*SELECT*]

options

*ACC_25NS ACC_100NS ACC_250NS ACC_1US ACC_2.5US ACC_10US
ACC_25US ACC_100US ACC_250US ACC_1MS ACC_2.5MS ACC_10MS
ACC_25MS ACC_100MS ACC_250MS ACC_1S ACC_10S ACC_10SPLUS
ACC_UNKNOWN*

usage

Clock accuracy range announced in master state.

default

ACC_UNKNOWN

ptpengine:utc_offset [*INT*]

usage

Underlying time source UTC offset announced in master state.

default

0

ptpengine:utc_offset_valid [*BOOLEAN*]

usage

Underlying time source UTC offset validity announced in master state.

default

N



ptpengine:time_traceable [*BOOLEAN*]

usage Underlying time source time traceability announced in master state.

default *N*

ptpengine:frequency_traceable [*BOOLEAN*]

usage Underlying time source frequency traceability announced in master state.

default *N*

ptpengine:ptp_timescale [*SELECT*]

options *PTP ARB*

usage Time scale announced in master state (with ARB, UTC properties are ignored by slaves). When clock class is set to 13 (application specific), this value is ignored and ARB is used.

default *ARB*

ptpengine:ptp_timesource [*SELECT*]

options *ATOMIC_CLOCK GPS TERRESTRIAL_RADIO PTP NTP HAND_SET OTHER INTERNAL_OSCILLATOR*

usage Time source announced in master state.

default *INTERNAL_OSCILLATOR*

ptpengine:clock_class [*INT: 0 .. 255*]

usage Clock class - announced in master state. Always 255 for slave-only. Minimum, maximum and default values are controlled by presets. If set to 13 (application specific time source), announced time scale is always set to ARB. This setting controls the states a PTP port can be in. If below 128, port will only be in MASTER or PASSIVE states (master only). If above 127, port will be in MASTER or SLAVE states.

default *255*

ptpengine:priority1 [*INT: 0 .. 248*]

usage Priority 1 announced in master state, used for Best Master Clock selection.

default *128*

ptpengine:priority2 [INT: 0 .. 248]

usage

Priority 2 announced in master state, used for Best Master Clock selection.

default

128

ptpengine:max_listen [INT: min: 1]

usage

Number of consecutive protocol resets to LISTENING before full network reset.

default

5

ptpengine:unicast_destinations [STRING]

usage

An IPv4 address or list of IPv4 addresses to be used as unicast destinations. When unicast negotiation (*ptpengine:unicast_negotiation*) is enabled, setting this is mandatory for slaves as they must be aware of which GMs to request messages from. When unicast negotiation is disabled, setting this is mandatory for GMs, as they must deliver messages to a pre-configured group of slaves.

default

[none]

ptpengine:unicast_domains [STRING]

usage

Specify PTP domain number for each configured unicast destination (*ptpengine:unicast_destinations*). This is only used by slave-only clocks using unicast destinations to allow for each master to be in a separate domain, such as with Telecom Profile. The number of entries should match the number of unicast destinations, otherwise unconfigured domains or domains set to 0 are set to domain configured in *ptpengine:domain*. The format is a comma, tab or space-separated list of 8-bit unsigned integers (0 .. 255).

default

[none]

ptpengine:unicast_local_preference [STRING]

usage

Specify a local preference for each configured unicast destination (*ptpengine:unicast_destinations*). This is only used by slave-only clocks using unicast destinations to allow for each master's BMC selection to be influenced locally by the slave, such as with Telecom Profile. The number of entries should match the number of unicast destinations, otherwise unconfigured preference is set to 0

(highest). The format is a comma, tab or space-separated list of 8-bit unsigned integers (0 .. 255).

default

[none]

ptpengine:unicast_peer_destination [STRING]

usage

When using IP unicast mode (*ptpengine:ip_mode=unicast*) and *Peer to Peer delay mechanism* (*ptpengine:delay_mechanism=P2P*), a peer unicast destination must be configured to request the peer delay from. Format is a single unicast IPv4 address.

default

[none]

ptpengine:management_enable [BOOLEAN]

usage

Enable handling of PTP management messages. Only GET messages are processed by default. See *ptpengine:management_set_enable*.

default

Y

ptpengine:management_set_enable [BOOLEAN]

usage

Accept SET and COMMAND management messages.

default

N

ptpengine:igmp_refresh [BOOLEAN]

usage

Send explicit IGMP joins between engine resets and periodically in master state.

default

Y

ptpengine:master_igmp_refresh_interval [INT: 0 .. 255]

usage

Periodic IGMP join interval (seconds) in master state when running IPv4 multicast: when set below 10 or when *ptpengine:igmp_refresh* is disabled, this setting has no effect.

default

60

ptpengine:multicast_ttl [INT: 1 .. 64]

usage

Multicast time to live for multicast PTP packets (ignored and set to 1 for peer to peer messages).

default

64

ptpengine:ip_dscp [INT: 0 .. 63]

usage

DiffServ CodePoint for packet prioritisation (decimal).
When set to zero, this option is not used. Use 46 for
Expedited Forwarding (0x2e).

default

0

ptpengine:sync_stat_filter_enable [BOOLEAN]

usage

Enable statistical filter for Sync messages

default

N

ptpengine:sync_stat_filter_type [SELECT]

options

none mean min max absmin absmax median

usage

Type of filter used for Sync message filtering:

none

no filtering - pass-through

mean

mean (average) - smooth results but influenced by
outliers

min

minimal value - useful for high packet delay
variation ("lucky packets")

max

maximal value - useful for testing worst case
scenarios

absmin

absolute minimum - value closest to zero. Also
useful for test purposes.

absmax

absolute maximum value farthest away from zero

median

median (middle value) - more robust than mean, not
influenced by outliers

default

min

ptpengine:sync_stat_filter_window [INT: 3 .. 128]

usage

Number of samples used for the Sync statistical filter

default

4

ptpengine:sync_stat_filter_window_type [SELECT]

options

sliding interval

usage

Sampling window behaviour for the Sync statistical filter:

sliding

sliding window - a value is output every time the filter runs, which can result in duplicates

interval

only output a value every n-th sample (full window)
- independent sampling periods

default

sliding

ptpengine:delay_stat_filter_enable [BOOLEAN]

usage

Enable statistical filter for Delay messages

default

N

ptpengine:delay_stat_filter_type [SELECT]

options

none mean min max absmin absmax median

usage

Type of filter used for Delay message filtering:

none

no filtering - pass-through

mean

mean (average) - smooth results but influenced by outliers

min

minimal value - useful for high packet delay variation ("lucky packets")

max

maximal value - useful for testing worst case scenarios

absmin

absolute minimum - value closest to zero. Also useful for test purposes.

absmax

absolute maximum value farthest away from zero

median

median (middle value) - more robust than mean, not influenced by outliers

default

min

ptpengine:delay_stat_filter_window [INT: 3 .. 128]

usage

Number of samples used for the Delay statistical filter

default

4

ptpengine:delay_stat_filter_window_type [SELECT]

options

sliding interval

usage

Sampling window behaviour for the Delay statistical filter:

sliding

sliding window - a value is output every time the filter runs, which can result in duplicates

interval

only output a value every n-th sample (full window)
- independent sampling periods

default

sliding

ptpengine:delay_outlier_filter_enable [BOOLEAN]

usage

Enable outlier filter for the Delay Response component in slave state

default

N

ptpengine:delay_outlier_filter_action [SELECT]

options

discard filter

usage

Delay Response outlier filter action. If set to 'filter', outliers are replaced with moving average.

default

discard

ptpengine:delay_outlier_filter_capacity [INT: 4 .. 60]

usage

Number of samples in the Delay Response outlier filter buffer

default

20

ptpengine:delay_outlier_filter_threshold [FLOAT: 0.001000 .. 1000.000000]

usage

Delay Response outlier filter threshold: multiplier for Peirce's maximum standard deviation. When set below 1.0, filter is tighter, when set above 1.0, filter is looser than standard Peirce's test.

default

1.000000

ptpengine:delay_outlier_filter_always_filter [BOOLEAN]

usage

Always run the Delay Response outlier filter, even if clock is being slewed at maximum rate

default

N



ptpengine:delay_outlier_filter_autotune_enable [*BOOLEAN*]

usage

Enable automatic threshold control for Delay Response outlier filter.

default

Y

ptpengine:delay_outlier_filter_autotune_minpercent [*INT: 0 .. 99*]

usage

Delay Response outlier filter autotune low watermark - minimum percentage of discarded samples in the update period before filter is tightened by the autotune step value

default

20

ptpengine:delay_outlier_filter_autotune_maxpercent [*INT: 1 .. 100*]

usage

Delay Response outlier filter autotune high watermark - maximum percentage of discarded samples in the update period before filter is loosened by the autotune step value

default

95

ptpengine:delay_outlier_filter_autotune_step [*FLOAT: 0.010000 .. 10.000000*]

usage

The value the Delay Response outlier filter threshold is increased or decreased by when auto-tuning

default

0.100000

ptpengine:delay_outlier_filter_autotune_minthreshold [*FLOAT: 0.010000 .. 10.000000*]

usage

Minimum Delay Response filter threshold value used when auto-tuning

default

0.100000

ptpengine:delay_outlier_filter_autotune_maxthreshold [*FLOAT: 0.010000 .. 10.000000*]

usage

Maximum Delay Response filter threshold value used when auto-tuning

default

5.000000

ptpengine:delay_outlier_weight [*FLOAT: 0.010000 .. 2.000000*]

usage

Delay Response outlier weight: if an outlier is detected, determines the amount of its deviation from mean that is used to build the standard deviation statistics and influence further outlier detection. When set to 1.0, the outlier is used as is.

default

1.000000

ptpengine:delay_outlier_filter_stepdetect_enable [*BOOLEAN*]

usage

Enable Delay Response filter step detection (delaySM) to block when certain level exceeded

default

N

ptpengine:delay_outlier_filter_stepdetect_threshold [*INT: 50000 .. 999999999*]

usage

Delay step detection threshold. Step detection is performed only when delaySM is below this threshold (nanoseconds)

default

1000000

ptpengine:delay_outlier_filter_stepdetect_level [*INT: 50000 .. 999999999*]

usage

Delay step level. When step detection enabled and operational, delaySM above this level (nanoseconds) is considered a clock step and updates are paused

default

500000

ptpengine:delay_outlier_filter_stepdetect_credit [*INT: 50 .. 1000*]

usage

Initial credit (number of samples) the Delay step detection filter can block for. When credit is exhausted, filter stops blocking. Credit is gradually restored (see *ptpengine:delay_outlier_filter_stepdetect_credit_increment*)

default

200

ptpengine:delay_outlier_filter_stepdetect_credit_increment [*INT: 1 .. 100*]

usage

Amount of credit for the Delay step detection filter restored every full sample window

default

10

ptpengine:sync_outlier_filter_enable [*BOOLEAN*]

usage

Enable outlier filter for the Sync component in slave state.

default
N

ptpengine:sync_outlier_filter_action [*SELECT*]

options
discard filter

usage
Sync outlier filter action. If set to 'filter', outliers are replaced with moving average.

default
discard

ptpengine:sync_outlier_filter_capacity [*INT: 4 .. 60*]

usage
Number of samples in the Sync outlier filter buffer.

default
20

ptpengine:sync_outlier_filter_threshold [*FLOAT: 0.001000 .. 1000.000000*]

usage
Sync outlier filter threshold: multiplier for the Peirce's maximum standard deviation. When set below 1.0, filter is tighter, when set above 1.0, filter is looser than standard Peirce's test.

default
1.000000

ptpengine:sync_outlier_filter_always_filter [*BOOLEAN*]

usage
Always run the Sync outlier filter, even if clock is being slewed at maximum rate

default
N

ptpengine:sync_outlier_filter_autotune_enable [*BOOLEAN*]

usage
Enable automatic threshold control for Sync outlier filter.

default
Y

ptpengine:sync_outlier_filter_autotune_minpercent [*INT: 0 .. 99*]

usage
Sync outlier filter autotune low watermark - minimum percentage of discarded samples in the update period before filter is tightened by the autotune step value

default
20

ptpengine:sync_outlier_filter_autotune_maxpercent [INT: 1 .. 100]

usage

Sync outlier filter autotune high watermark - maximum percentage of discarded samples in the update period before filter is loosened by the autotune step value

default

95

ptpengine:sync_outlier_filter_autotune_step [FLOAT: 0.010000 .. 10.000000]

usage

The value the Sync outlier filter threshold is increased or decreased by when auto-tuning

default

0.100000

ptpengine:sync_outlier_filter_autotune_minthreshold [FLOAT: 0.010000 .. 10.000000]

usage

Minimum Sync filter threshold value used when auto-tuning

default

0.100000

ptpengine:sync_outlier_filter_autotune_maxthreshold [FLOAT: 0.010000 .. 10.000000]

usage

Maximum Sync filter threshold value used when auto-tuning

default

5.000000

ptpengine:sync_outlier_weight [FLOAT: 0.010000 .. 2.000000]

usage

Sync outlier weight: if an outlier is detected, this value determines the amount of its deviation from mean that is used to build the standard deviation statistics and influence further outlier detection.

When set to 1.0, the outlier is used as is

default

1.000000

ptpengine:sync_outlier_filter_stepdetect_enable [BOOLEAN]

usage

Enable Sync filter step detection (delayMS) to block when certain level exceeded

default

N

ptpengine:sync_outlier_filter_stepdetect_threshold [INT: 50000 .. 999999999]

usage

Sync step detection threshold. Step detection is performed only when delayMS is below this threshold (nanoseconds)

default

1000000

ptpengine:sync_outlier_filter_stepdetect_level [*INT*: 50000 .. 999999999]

usage

Sync step level. When step detection enabled and operational, delayMS above this level (nanoseconds) is considered a clock step and updates are paused

default

500000

ptpengine:sync_outlier_filter_stepdetect_credit [*INT*: 50 .. 1000]

usage

Initial credit (number of samples) the Sync step detection filter can block for. When credit is exhausted, filter stops blocking. Credit is gradually restored (see *ptpengine:sync_outlier_filter_stepdetect_credit_increment*)

default

200

ptpengine:sync_outlier_filter_stepdetect_credit_increment [*INT*: 1 .. 100]

usage

Amount of credit for the Sync step detection filter restored every full sample window

default

10

ptpengine:sync_sequence_checking [*BOOLEAN*]

usage

When enabled, Sync messages will only be accepted if sequence ID is increasing. **note:** This can cause the slave to temporarily lock up if GM restarts before announce timeout, so this is limited to 50 consecutive sequence errors. Alternatively, *ptpengine:clock_update_timeout* can be used to reset the slave beforehand.

default

N

ptpengine:clock_update_timeout [*INT*: 0 .. 3600]

usage

If set to non-zero, time (seconds) before slave is reset back into PTP_LISTENING, if there were no clock updates. This is useful for situations where slave is in SLAVE state (receiving Announce) but is not receiving or not accepting Sync messages.

default

0

ptpengine:calibration_delay [*INT*: 0 .. 300]

usage

Delay between moving to slave state and enabling clock updates (seconds). This allows mean path delay to stabilise before starting clock updates. Activated when going into slave state and during slave's GM failover. 0 - not used.

default

0

ptpengine:idle_timeout [INT: 10 .. 3600]

usage

PTP idle timeout (seconds): if PTPd is in SLAVE state and there have been no clock updates for this amount of time, PTPd releases clock control.

default

0

ptpengine:panic_mode [BOOLEAN]

usage

Enable panic mode: when offset from master is above 1 second, stop updating the clock for a period of time and then step the clock if offset remains above 1 second.

default

N

ptpengine:panic_mode_duration [INT: 1 .. 60]

usage

Duration (minutes) of the panic mode period (no clock updates) when offset above 1 second detected.

default

2

ptpengine:panic_mode_release_clock [BOOLEAN]

usage

When entering panic mode, release clock control while panic mode lasts. If not set, PTP will hold clock control during panic mode. If set together with ntpengine:* configured, this will fail over to NTP.

default

N

ptpengine:panic_mode_exit_threshold [INT: 0 .. 999999999]

usage

Do not exit panic mode until offset drops below this value (nanoseconds). 0 = not used.

default

0

ptpengine:pid_as_clock_identity [BOOLEAN]

usage

Use PTPd's process ID as the middle part of the PTP clock ID - useful for running multiple instances.

default

N

ptpengine:ntp_failover [*BOOLEAN*]

usage

Fail over to NTP when PTP time sync not available - requires ntpengine:enabled, but does not require the rest of NTP configuration: will warn instead of failing over if cannot control ntpd.

default

N

ptpengine:ntp_failover_timeout [*INT: 0 .. 1800*]

usage

NTP failover timeout in seconds: time between PTP slave going into LISTENING state, and failing over to NTP. 0 = fail over immediately. This setting controls the time provider election hold time.

default

60

ptpengine:prefer_ntp [*BOOLEAN*]

usage

Prefer NTP time synchronisation. Only use PTP when NTP not available. Could be used when NTP runs with a local GPS receiver or another hardware reference.

default

N

ptpengine:panic_mode_ntp [*BOOLEAN*]

usage

Deprecated as of 2.3.1, but still supported: see *ptpengine:panic_mode_release_clock*.

default

N

ptpengine:sigusr2_clears_counters [*BOOLEAN*]

usage

Clear counters after dumping all counter values on SIGUSR2.

default

N

ptpengine:timing_acl_permit [*STRING*]

usage

Permit access control list for timing and signaling messages. Format is a series of network prefixes and/or IP addresses separated by commas, spaces, tabs or semicolons. Accepted format is CIDR notation (a.b.c.d/mm), single IP address (a.b.c.d), or full network/mask (a.b.c.d/m.m.m.m). Shortcuts can be used: 172.16/12 is expanded to 172.16.0.0/12; 192.168/255.255 is expanded to

192.168.0.0/255.255.0.0, etc. The match is performed on the source IP address of the incoming messages. IP access lists are only supported when using the IP transport.

default

[none]

ptpengine:timing_acl_deny *[STRING]*

usage

Deny access control list for timing and signaling messages. Format is a series of network prefixes and/or IP addresses separated by commas, spaces, tabs or semicolons. Accepted format is CIDR notation (a.b.c.d/mm), single IP address (a.b.c.d), or full network/mask (a.b.c.d/m.m.m.m). Shortcuts can be used: 172.16/12 is expanded to 172.16.0.0/12; 192.168/255.255 is expanded to 192.168.0.0/255.255.0.0, etc. The match is performed on the source IP address of the incoming messages. IP access lists are only supported when using the IP transport.

default

[none]

ptpengine:management_acl_permit *[STRING]*

usage

Permit access control list for management messages. Format is a series of network prefixes and/or IP addresses separated by commas, spaces, tabs or semicolons. Accepted format is CIDR notation (a.b.c.d/mm), single IP address (a.b.c.d), or full network/mask (a.b.c.d/m.m.m.m). Shortcuts can be used: 172.16/12 is expanded to 172.16.0.0/12; 192.168/255.255 is expanded to 192.168.0.0/255.255.0.0, etc. The match is performed on the source IP address of the incoming messages. IP access lists are only supported when using the IP transport.

default

[none]

ptpengine:management_acl_deny *[STRING]*

usage

Deny access control list for management messages. Format is a series of network prefixes and/or IP addresses separated by commas, spaces, tabs or semicolons. Accepted format is CIDR notation (a.b.c.d/mm), single IP address (a.b.c.d), or full network/mask (a.b.c.d/m.m.m.m). Shortcuts can be used: 172.16/12 is expanded to 172.16.0.0/12; 192.168/255.255 is expanded to 192.168.0.0/255.255.0.0, etc. The match is performed on the source IP address of the incoming messages. IP access lists are only supported when using the IP transport.

default

[none]

ptpengine:timing_acl_order *[SELECT]*

options

permit-deny deny-permit

usage

Order in which permit and deny access lists are evaluated for timing and signaling messages, the evaluation process

is the same as for Apache httpd. **See:**
http://httpd.apache.org/docs/current/mod/mod_access_compat.html#order

default

deny-permit

ptpengine:management_acl_order [*SELECT*]

options

permit-deny deny-permit

usage

Order in which permit and deny access lists are evaluated for management messages, the evaluation process is the same as for Apache httpd. **See:**
http://httpd.apache.org/docs/current/mod/mod_access_compat.html#order

default

deny-permit

clock:no_adjust [*BOOLEAN*]

usage

Do not adjust the clock.

default

N

clock:no_reset [*BOOLEAN*]

usage

Do not reset the clock - only slew.

default

N

clock:step_startup_force [*BOOLEAN*]

usage

Force clock step on first sync after startup regardless of offset and *clock:no_reset*

default

N

clock:step_startup [*BOOLEAN*]

usage

Step clock on startup only if offset >= 1 second, ignoring panic mode and *clock:no_reset*

default

N

clock:set_rtc_on_step [*BOOLEAN*]

usage

Attempt setting the RTC when stepping clock (Linux only - FreeBSD does this for us. **WARNING:** this will always set the RTC to OS clock time, regardless of time zones, so this assumes that RTC runs in UTC or otherwise in the same

timescale as PTP. True at least on most single-boot x86 Linux systems.

default

N

clock:drift_handling [*SELECT*]

options

reset preserve file

usage

Observed drift handling method between servo restarts:

reset

set to zero (not recommended)

preserve

use kernel value

file

load/save to drift file on startup/shutdown, use kernel value inbetween. To specify drift file, use the **clock:drift_file** setting.

default

preserve

clock:drift_file [*STRING*]

usage

Specify drift file

default

/etc/ptpd2_kernelclock.drift

clock:leap_seconds_file [*STRING*]

usage

Specify leap second file location (up to date version can be downloaded from: <http://www.ietf.org/timezones/data/leap-seconds.list>). When configured, PTP master will use data from this file to announce leap flags and UTC offset, overriding OS information, and PTP slave will use data from this file as well as information supplied by the GM. If configured, this file is always reloaded on configuration reload (SIGHUP), reloaded on clock step and reloaded after a leap second event to ensure the information is up to date. As of ptpd 2.3.1, the file is bundled with ptpd and is installed into (prefix)/share/ptpd/leap-seconds.list.ddMMyyyy where ddMMyyyy is the leap seconds file expiry date.

default

[none]

clock:leap_second_pause_period [*INT*: 5 .. 600]

usage

Time (seconds) before and after midnight that clock updates should be suspended for during a leap second event. The total duration of the pause is twice the configured duration. Clock updates are suspended when there is a leap second event pending and time to midnight is less than or equal to this value and resumed no earlier than this value after midnight. Clock updates are resumed in a controlled manner - after a control message, such as PTP announce.

This ensures that the updated UTC offset is received before any further clock updates.

default

5

clock:leap_second_notice_period [*INT*: 3600 .. 86400]

usage

Time (seconds) before midnight that PTPd starts announcing the leap second if it's running as master. The IEEE 1588 standard suggests 12 hours notice and this is the default, but it may be changed to allow more flexibility.

default

43200

clock:leap_second_handling [*SELECT*]

options

accept ignore step smear

usage

Clock sync behaviour during leap second events:

accept

Inform OS kernel about the leap second and let the kernel insert or delete the leap second

ignore

Do not inform the kernel - this ends with a +/-1-second offset which is then slewed back down

step

Do not inform the kernel and step the clock immediately after the leap second event

smear

Gradually introduce an extra offset over a period of time before the leap second event, which accumulates to +/-1 second (see *clock:leap_second_smear_period*). Once the clock stabilises, this results in a clock frequency shift which is taken off after the event. Once the leap second event is over, the extra offset is also removed and time is back in line with master time.

default

accept

clock:leap_second_smear_period [*INT*: 3600 .. 86400]

usage

When *clock:leap_second_handling* is set to *smear*, this setting defines the period (in seconds) before the leap second event, over which the leap second offset is gradually added. Example: when set to 24 hours (86400), an extra +/-11.5 microseconds is added every second (11.5 ppm clock frequency offset).

default

86400

clock:max_offset_ppm [*INT*: 500 .. 1000]

usage

Maximum absolute frequency shift which can be applied to the clock servo when slewing the clock. Expressed in parts per million (1 ppm = shift of 1 us per second. Values above 512 will use the tick duration correction to allow even faster slewing. Default maximum is 512 without using tick.

default

500

servo:delayfilter_stiffness [*INT*]

usage

Mean Path Delay filter stiffness.

default

6

servo:kp [*FLOAT*: min: 0.000001]

usage

Clock servo PI controller proportional component gain (kP).

default

0.100000

servo:ki [*FLOAT*: min: 0.000001]

usage

Clock servo PI controller integral component gain (kI).

default

0.001000

servo:dt_method [*SELECT*]

options

none constant measured

usage

How servo update interval (delta t) is calculated:

none

servo not corrected for update interval (dt always 1),

constant

constant value (target servo update rate) - sync interval for PTP,

measured

servo measures how often it's updated and uses this interval.

default

constant

servo:dt_max [*FLOAT*: 1.500000 .. 100.000000]

usage

Maximum servo update interval (delta t) when using measured servo update interval (*servo:dt_method* = **measured**), specified as sync interval multiplier

default

5.000000

servo:stability_detection [*BOOLEAN*]

usage

Enable clock synchronisation servo stability detection (based on standard deviation of the observed drift value) - drift will be saved to drift file / cached when considered stable, also clock stability status will be logged.

default

N

servo:stability_threshold [*FLOAT: 1.000000 .. 10000.000000*]

usage

Specify the observed drift standard deviation threshold in parts per billion (ppb) - if standard deviation is within the threshold, servo is considered stable.

default

5.000000

servo:stability_period [*INT: 1 .. 100*]

usage

Specify for how many statistics update intervals the observed drift standard deviation has to stay within threshold to be considered stable.

default

3

servo:stability_timeout [*INT: 1 .. 60*]

usage

Specify after how many minutes without stabilisation servo is considered unstable. Assists with logging servo stability information and allows one to preserve observed drift if servo cannot stabilise.

default

10

servo:max_delay [*INT: 0 .. 999999999*]

usage

Do not accept master to slave delay (delayMS - from Sync message) or slave to master delay (delaySM - from Delay Response message) if greater than this value (nanoseconds). 0 = not used.

default

0

servo:max_delay_max_rejected [*INT: min: 0*]

usage

Maximum number of consecutive rejected delay measurements exceeding the maxDelay threshold (*servo:max_delay*), before slave is reset. 0 = not checked.

default

0

servo:max_delay_stable_only [*BOOLEAN*]

usage

If **servo:max_delay** is set, perform the check only if clock servo has stabilised.

default

N

servo:max_offset [*INT: 0 .. 999999999*]

usage

Do not reset the clock if offset from master is greater than this value (nanoseconds). 0 = not used.

default

0

global:enable_snmp [*BOOLEAN*]

usage

Enable SNMP agent (if compiled with PTPD_SNMP).

default

N

global:use_syslog [*BOOLEAN*]

usage

Send log messages to syslog. Disabling this sends all messages to stdout (or specified log file).

default

N

global:lock_file [*STRING*]

usage

Lock file location

default

[none]

global:auto_lockfile [*BOOLEAN*]

usage

Use mode specific and interface specific lock file (overrides **global:lock_file**).

default

N

global:lock_directory [*STRING*]

usage

Lock file directory: used with automatic mode-specific lock files, also used when no lock file is specified. When lock file is specified, it's expected to be an absolute path.

default

/var/run

global:ignore_lock [*BOOLEAN*]

usage
Skip lock file checking and locking.
default
N

global:quality_file [*STRING*]

usage
File used to record data about sync packets. Enables recording when set.
default
[none]

global:quality_file_max_size [*INT: min: 0*]

usage
Maximum sync packet record file size (in kB) - file will be truncated if size exceeds the limit. 0 - no limit.
default
0

global:quality_file_max_files [*INT: 0 .. 100*]

usage
Enable log rotation of the sync packet record file up to n files. 0 - do not rotate.
default
0

global:quality_file_truncate [*BOOLEAN*]

usage
Truncate the sync packet record file every time it is (re) opened: startup and SIGHUP.
default
N

global:status_file [*STRING*]

usage
File used to log ptpd2 status information.
default
/var/run/ptpd2.status

global:log_status [*BOOLEAN*]

usage
Enable / disable writing status information to file.
default
N

global:status_update_interval [*INT: 1 .. 30*]

usage
Status file update interval in seconds.

default
1

global:log_file [*STRING*]

usage
Specify log file path (event log). Setting this enables logging to file.

default
[none]

global:log_file_max_size [*INT*: min: 0]

usage
Maximum log file size (in kB) - log file will be truncated if size exceeds the limit. 0 - no limit.

default
0

global:log_file_max_files [*INT*: 0 .. 100]

usage
Enable log rotation of the sync packet record file up to n files. 0 - do not rotate.

default
0

global:log_file_truncate [*BOOLEAN*]

usage
Truncate the log file every time it is (re) opened: startup and SIGHUP.

default
N

global:log_level [*SELECT*]

options
LOG_ERR LOG_WARNING LOG_NOTICE LOG_INFO LOG_ALL

usage
Specify log level (only messages at this priority or higher will be logged). The minimal level is LOG_ERR. LOG_ALL enables debug output if compiled with RUNTIME_DEBUG.

default
LOG_ALL

global:statistics_file [*STRING*]

usage
Specify statistics log file path. Setting this enables logging of statistics, but can be overridden with **global:log_statistics**.

default
[none]

global:statistics_log_interval [*INT*: min: 0]

usage
Log timing statistics every n seconds for Sync and Delay messages (0 - log all).

default
0

global:statistics_file_max_size [*INT*: min: 0]

usage
Maximum statistics log file size (in kB) - log file will be truncated if size exceeds the limit. 0 - no limit.

default
0

global:statistics_file_max_files [*INT*: 0 .. 100]

usage
Enable log rotation of the statistics file up to n files. 0 - do not rotate.

default
0

global:statistics_file_truncate [*BOOLEAN*]

usage
Truncate the statistics file every time it is (re) opened: startup and SIGHUP.

default
N

global:dump_packets [*BOOLEAN*]

usage
Dump the contents of every PTP packet.

default
N

global:verbose_foreground [*BOOLEAN*]

usage
Run in foreground with statistics and all messages logged to stdout. Overrides log file and statistics file settings and disables syslog.

default
N

global:foreground [*BOOLEAN*]

usage

Run in foreground - ignored when `global:verbose_foreground` is set.

default

N

`global:log_statistics` [*BOOLEAN*]

usage

Log timing statistics for every PTP packet received. Output is in CSV format and field headers are always printed when starting or refreshing the statistics log.

default

N

`global:statistics_timestamp_format` [*SELECT*]

options

datetime unix both

usage

Timestamp format used when logging timing statistics (when `global:log_statistics` is enabled):

datetime

Formatted date and time: `YYYY-MM-DD hh:mm:ss.uuuuuu`

unix

Unix timestamp with nanoseconds: `s.ns`

both

Formatted date and time followed by unix timestamp (adds one extra field to the log)

default

datetime

`global:periodic_updates` [*BOOLEAN*]

usage

Log a status update every time statistics are updated (`global:statistics_update_interval`). This update is written to the main log target. Status updates are logged even if ptpd is configured without support for statistics.

default

N

`global:cpuaffinity_cpucore` [*INT*: -1 .. 255]

usage

Bind ptpd2 process to a selected CPU core number. 0 = first CPU core, etc. -1 = do not bind to a single core.

default

0

`global:statistics_update_interval` [*INT*: 1 .. 60]

usage

Clock synchronisation statistics update interval in seconds. Also controls how often periodic status information is logged (when using `global:statistics_update_interval`).

default

30

global:timingdomain_election_delay [INT: 0 .. 3600]

usage

Delay (seconds) before releasing a time service (NTP or PTP) and electing a new one to control a clock. 0 = elect immediately

default

15

ntpengine:enabled [BOOLEAN]

usage

Enable NTPd integration

default

N

ntpengine:control_enabled [BOOLEAN]

usage

Enable control over local NTPd daemon

default

N

ntpengine:check_interval [INT: 5 .. 600]

usage

NTP control check interval in seconds

default

15

ntpengine:key_id [INT: 0 .. 65535]

usage

NTP key number - must be configured as a trusted control key in ntp.conf, and be non-zero for the ntpengine:control_enabled setting to take effect.

default

0

ntpengine:key [STRING]

usage

NTP key (plain text, max. 20 characters) - must match the key configured in ntpd's keys file, and must be non-zero for the ntpengine:control_enabled setting to take effect.

default

[none]

BUGS

Configuration file support has only been introduced in version 2.3. There may still be some inconsistencies in the way some settings are parsed and while order should not make any difference, for some complex behaviours it may still be the case.

Please report any bugs using the bug tracker on the SourceForge page:

<http://sourceforge.net/projects/ptpd/>

SEE ALSO

ptpd2(8)

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